

Compare Theoretical and Experimental Probability

Step 9

Upper Elementary
(Grades 4–5)



Today I am learning to compare theoretical (expected) probability with experimental (actual) probability.

1. What Is Theoretical Probability?

Theoretical probability is what we expect to happen based on equal chances. Write the theoretical probability for each event.

- a) Rolling a 3 on a number cube.



Theoretical probability: $\frac{1}{6}$

- b) Spinning a spinner with 6 equal sections. Landing on blue.



Theoretical probability: $\frac{1}{6}$

- c) Picking a yellow counter from the bag.



Theoretical probability: $\frac{1}{5}$

- d) Flipping a coin. Getting tails.



Theoretical probability: $\frac{1}{2}$

2. Do the Experiment and Record Results

Do the experiment. Record how many times each outcome happened.

- a) Roll a number cube 30 times.



Number	1	2	3	4	5	6
Tally						
Total						

- b) Spin the spinner 30 times.



Colour	Red	Blue	Green	Yellow	Orange	Purple
Tally						
Total						

- c) Pick a counter from the bag 30 times.

Colour	Red	Blue	Yellow
Tally			
Total			

- d) Flip a coin 30 times.



Outcome	Heads	Tails
Tally		
Total		

3. Compare Theoretical and Experimental Probabilities

Use your results from Question 2. Complete the table.

- a) Number cube (30 rolls)

Outcome	Theoretical Probability	Experimental Probability	Difference (Experimental – Theoretical)
1			
2			
3			
4			
5			
6			

- b) Spinner (30 spins)

Colour	Theoretical Probability	Experimental Probability	Difference (Experimental – Theoretical)
Red			
Blue			
Green			
Yellow			
Orange			
Purple			

- c) Counters in bag (30 picks)

Colour	Theoretical Probability	Experimental Probability	Difference (Experimental – Theoretical)
Red			
Blue			
Green			
Yellow			

- d) Coin (30 flips)

Outcome	Theoretical Probability	Experimental Probability	Difference (Experimental – Theoretical)
Heads			
Tails			

4. Analyse the Results

Look at your tables in Question 3. Answer the questions.

- a) Which outcome was closest to the theoretical probability in each experiment?

- Number cube: _____
- Spinner: _____
- Counters in bag: _____
- Coin: _____



- b) Which outcome was furthest from the theoretical probability in each experiment?

- Number cube: _____
- Spinner: _____
- Counters in bag: _____
- Coin: _____



- c) Why might the experimental probability be different from the theoretical probability?



- d) If you repeated the experiment 100 times instead of 30, what might happen to the experimental probabilities? Why?



5. Make a Prediction

Predict the experimental probability if each experiment is repeated 60 times. Write your prediction as a fraction.

- a)  My prediction: $\frac{\quad}{\quad}$



- c)  Picking a blue counter from the bag $\frac{\quad}{\quad}$

- b)  Landing on green on the spinner $\frac{\quad}{\quad}$



- d)  Getting tails on a coin $\frac{\quad}{\quad}$

7. Challenge

Design your own experiment using a spinner with 8 equal sections.

Do the experiment 40 times and record your results.

Compare your experimental probabilities with the theoretical probabilities.



Colour	Theoretical Probability	Experimental Probability (out of 40)	Difference (Exp – Theo)
Red			
Orange			
Yellow			
Green			
Blue			
Purple			
Black			

6. Real-Life Connection

A basketball team has a 60% chance of winning their next game based on past results.

- What is the theoretical probability of winning? _____
- They play the game 10 times and win 4 times. What is the experimental probability of winning? _____
- Is the experimental probability higher, lower or the same as the theoretical probability? _____
- If they played 100 games, why might the experimental probability change? _____



8. Think and Explain

- How do theoretical and experimental probability help us understand chance? _____
- Can an event ever have an experimental probability of 0 if the theoretical probability is not 0? Explain. _____
- How does the number of trials affect how close experimental probability gets to theoretical probability? _____



I can...

- ☐ find theoretical probability for simple events.
- ☐ conduct experiments and record results.
- ☐ compare theoretical and experimental probabilities.
- ☐ explain why differences occur.

theoretical (expected)

experimental (actual)

Chance Words

impossible

unlikely

equally likely

likely

fair

likely

unlikely

certain

0%

less than 50%

about 50%

more than 50%

100%



Great job!

You are comparing chance like a mathematician!